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March 2026

Identifying Misinformation Spreaders: Challenges and Model Architectures

Bachelor's Thesis

to confer the academic degree of

Bachelor of Science

in the Bachelor's Program

Artificial Intelligence

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Abstract

With the Internet's growing role as the primary medium for information dissemination, traditional models of news consumption have shifted, with social media platforms emerging as dominant sources for users. The ease of sharing information on these platforms has led to a significant increase in misinformation creation and its propagation. The propagation of false or misleading news not only contributes to psychological distress and emotional overload but also has far-reaching negative consequences on individuals and society as a whole. As a result, there is growing interest in developing models to identify potential spreaders of misinformation. However, detecting misinformation spreaders presents unique challenges that traditional approaches struggle to address. Misinformation is produced with the intent to mislead users and can include partially correct content, references to accurate information taken out of context, or accounts of events that lack verifiable sources at the time of reporting. These issues require a model capable of addressing these limitations. In this report, we discuss the challenges associated with detecting misinformation spreaders and propose two models with different feature sets and architectures, including a lightweight variant. The models combine content-based features with graph-based representations for more fine-grained analysis. Evaluated against traditional attribute-based and state-of-the-art baselines on a benchmark dataset with a single temporal split, the primary model achieves the best Matthews correlation coefficient and F1-score, while the lightweight variant achieves the second-best Matthews score.

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Chapter 1

Introduction

The rise of online social media platforms has profoundly transformed social interactions and information consumption patterns. The reliance on these platforms is skyrocketing with 86% users in 2024 reporting that they had accessed news via a smartphone, computer, or tablet at least occasionally[1], and 54% stating that they had frequently consumed news from social media[2]. The convenience and simplicity of sharing news online, combined with the lack of strict regulations regarding misinformation spread, and the significantly faster spread of misinformation compared with that of accurate news[3], has contributed to a drastically rise of the number of misinformation spreaders. This phenomenon poses serious societal risks as evidenced during the COVID-19 pandemic, when the exposure to incorrect information or rumors was directly associated with increased anxiety levels among society [21]. To address this issue, binary classification models that differentiate misinformation spreaders from regular users have been developed as a potential countermeasure to mitigate the dissemination of misinformation.

Identifying misinformation spreaders presents several challenges:

1. Misinformation is often intentionally written to deceive users into believing it. For example, the Internet Research Agency in St. Petersburg conducted an information operation to influence the 2016 United States election. This operation was aimed, in part, to influence political views in the United States leading up to the 2016 election [31].

2. Some false or misleading content might be partially correct or reference legitimate facts or information[4].

3. Immediately verifying misinformation can be difficult, as they might be referring to recent events for which credible sources may not yet be available.

4. The credibility of certain information may remain unverifiable. As a result, solely relying on content for identifying misinformation has proven ineffective. To overcome these limitations, novel models that leverage either interaction graphs or text-based feature extraction techniques to analyze writing style (e.g., [8, 11, 23, 25, 29]) have been introduced. These approaches mimic human decision-making when contextual knowledge is missing and detection focuses on linguistic patterns and references rather than the content itself. Given the diverse range of topics, styles, and intentions underlying misinformation, effective detection models must account for these variations.

[1] https://www.pewresearch.org/journalism/fact-sheet/news-platform-fact-sheet/
[2] https://www.pewresearch.org/journalism/fact-sheet/social-media-and-news-fact-sheet/
[3] https://news.mit.edu/2018/study-twitter-false-news-travels-faster-true-stories-0308/
[4] https://de.in-mind.org/blog/post/fake-news-entkraeften-mit-dem-truth-sandwich-der-neue-ansatz-unter-der-lupe

1.1 Problem Definition

Let u be a user who engages in a public activity on some social media platform. Each user account contains metadata, including the user's name, description, and profile picture. A user u shared n publications (e.g., tweets), represented as $c = [a_1, a_2, ..., a_n]$, where each publication a_i includes text and might also contain additional content such as: hyperlinks, images, timestamp, mentions to other users or relations to another a_k (e.g., through replies, shares or quotes). Let U refer to a list of users $[u_1, u_2, ..., u_l]$.

The task of detecting misinformation spreaders is defined as a function F that takes a user u as input and produces a binary classification $\mathtt{result} \rightarrow \{0, 1\}$, where 1 indicates that u is a misinformation spreader and 0 otherwise. In this setting, the objective is to predict the label of users who were not observed during training. Consequently, the training and test sets contain disjoint users, and the model must generalize to previously unseen users based on their observable behavior and content.

The ground label is determined by comparing p, the proportion of a user's shared publications that contain misinformation, to a predefined threshold k. For example, a user u is labeled a misinformation spreader if $p \geq k$, where $k = 0.5$, meaning that a user is classified as a misinformation spreader if at least 50% of their shared content consists of misinformation. Another possible approach is multiclass classification. Multiple k values are used to partition the task into predicting the class to which a user belongs. Assuming we have m thresholds, the classes are defined as $[0, k_1), ..., (k_m, 1]$.

For the purpose of this study, we define misinformation as any piece of information that is either partially or entirely composed of fake or misleading content. This definition is agnostic to the intent behind the misinformation, meaning it does not account for whether the spread was deliberate, accidental, or the result of satire or sarcasm. As the context in which a piece of misinformation was used is not considered, this definition is closely related to the problem of rumor detection in social media.

1.2 Organization of the Document

The rest of this document is organized as follows.

Chapter 2 provides a comprehensive review of state-of-the-art models for misinformation spreader detection, covering content-based approaches, graph-based methods, hybrid structural models, and temporal encoding techniques. This chapter highlights the strengths and limitations of existing approaches.

Chapter 3 provides an overview of the datasets used in this study. Furthermore, it details the extracted linguistic features, including a brief description of each feature set, followed by an exploratory analysis to assess their relative discriminative power.

Chapter 4 introduces the proposed modeling framework. It presents the four components of the proposed model, along with implementation details describing how the data is combined to produce the final prediction. It discusses three graphs incorporated into the model: the User-with-User Similarity Graph (UUSG), the Publication-with-Publication Interaction Graph (PPIG), and the User-with-User Interaction Graph (UUIG). Furthermore, it describes how the temporal information is integrated into the framework. The chapter concludes with a lightweight model, the Similarity-Interaction-United Model (SIUM), along with its associated pipeline.

Chapter 5 provides the experimental evaluation of the proposed approach. It outlines experimental settings for reproducibility, baseline models, and implementation specifics. This chapter examines the following three research questions in detail, supported by comprehensive experimental evaluation.

1. **RQ1.** How does the proposed model compare with state-of-the-art techniques in predicting the future behavior of users?
2. **RQ2.** What is the contribution of each component of the proposed pipeline to the overall performance of the model?
3. **RQ3.** How does the proposed model perform in a multi-class prediction setting?

Finally, **Chapter 6** concludes the research by summarizing the main contributions, outlining the limitations, and providing directions for future research.

Chapter 2

Related work

The detection of misinformation remains an open research question that has gained increased attention in recent years due to massive societal consequences. Various techniques have been proposed to tackle this problem [30]. Early techniques focused on leveraging linguistic characteristics to distinguish misinformation spreaders [16]. However, since misinformation is often designed in a way to imitate legitimate news articles, distinguishing between misinformation and factual reporting remains a complex task. To address this challenge, more advanced architectures have been introduced, designed to handle the inherent difficulty of classifying content where both misinformation and factual news share similar linguistic structures and communicative intent.

This section discusses several existing model architectures, categorized into three main groups: Content-Based [7, 8, 15, 29], Graph-Based [12, 23, 26], and Hybrid [11, 25, 33] Misinformation Spreader Detection, which forms the basis of our proposed model. Additionally, we discuss the role of temporal encoding, which has been incorporated in some recent studies to enhance detection [7, 15, 23].

2.1 Content-Based Misinformation Spreader Detection

Content-based approaches primarily focus on the textual content of users' publications while disregarding their social interactions. These models extract features such as tone, writing style, readability, personality, and other textual metrics to assess whether a user is likely to spread misinformation. Content-based models typically employ Feedforward Neural Networks (FFNs) or Random Forests (RFs) to predict user behavior from the extracted features. For example, the model presented by Shrestha and Spezzano [29] incorporates various linguistic and behavioral features, including writing style, user emotions, user popularity metrics, personality traits, and emotions, alongside several manually engineered features. Then, the features are passed through an RF classifier, achieving an average precision ranging between 0.80 and 0.99 across the evaluated datasets. Another example is the *CheckerOrSpreader* model proposed by Giachanou et al. [8]. This model incorporates a psycholinguistic component(psychometric and linguistic style habits that were derived from the textual content of each publication, such as sentiment, emotions, linguistic patterns, personality traits, communication style, and readability), enriching it with an embedding layer to capture additional semantic meaning. The model avoids the use of manually engineered features, as such features may limit the model's ability to capture the complexity of misinformation.

Content-based approaches suffer from several limitations. First, by focusing primarily on textual features, they largely ignore the social and relational context in which misinformation is produced and disseminated, even though such context is crucial for capturing

coordinated behavior, diffusion patterns, and the roles users play in spreading information. In practice, benign users and misinformation spreaders may exhibit very similar writing styles or topical preferences, making them difficult to distinguish based on content alone, especially when messages are neutral, reused, or adapted to evade detection. Second, many of these methods rely on handcrafted or surface-level features, which may capture stylistic regularities but struggle to model collective and strategic behaviors, such as coordinated amplification or community-driven propagation. Finally, content-based models are often sensitive to domain shifts and changes in writing style, which can limit their generalization ability across topics, platforms, or time periods.

2.2 Graph-Based Misinformation Spreader Detection

Misinformation spreaders often engage in conversation threads and interact with other users, making relational information a crucial factor for understanding their behavior. Graph-based approaches exploit these interactions by constructing user interaction or similarity graphs and applying Graph Neural Networks (GNNs) to predict whether a user is a misinformation spreader. Several recent methods leverage GNNs to capture relational dependencies and propagation patterns in social media networks.

For example, Rath et al. [26] model the graph structure as a medium for misinformation dissemination and incorporate the concept of interpersonal trust [27], a graph-based property intended to capture a node's trustworthiness. These trust scores are used as weights for neighborhood sampling and feature aggregation. However, the interpersonal trust model is not publicly available, which limits reproducibility and reduces the transparency and explanatory potential of the approach.

Another example is the model proposed by Hu et al. [12], which incorporates a multi-depth input matrix defined as multiple powers of the normalized adjacency matrix. The model applies graph convolution to learn node representations based on textual features—obtained from fixed-length embeddings of user publications and historical credibility indicators derived from fact-checking outcomes in political speech data. The resulting representations are fused through an attention mechanism and fed into a classifier. Notably, this approach is designed for a politics-specific setting, which may limit its applicability to other domains.

Despite their advantages, graph-based approaches also exhibit important limitations. Many methods rely on domain-specific assumptions, handcrafted graph constructions, or auxiliary signals such as trust or credit history, which may not be available or reliable across different platforms and contexts. Moreover, several models are tightly coupled to specific datasets or domains (e.g., political discourse), raising concerns about their generalization to other topics and scenarios. Finally, while graph-based models can capture relational structure, they often come with increased computational complexity and reduced interpretability, making it harder to deploy them in large-scale or real-world settings and to explain their predictions to end users.

2.3 Hybrid Structure Misinformation Spreader Detection

Hybrid models integrate graph-based representations with content-derived features, aiming to combine the strengths of both approaches. While this integration typically increases computational complexity, it can substantially improve detection performance. In these

models, graph-embedded features learned by GNNs or graph embedding methods are combined with content or user-level features and subsequently fed into a classifier, often a Feedforward Neural Network (FNN).

For instance, Hamdi et al. [11] employ a Node2vec [10] embedding process, combined with a Skip-gram approach, to generate node embeddings from graph structures. These embeddings are concatenated with baseline user-based features to enrich the model with additional information, resulting in significant improvements over several state-of-the-art methods. This multimodal fusion strategy captures both network-driven and content-driven indicators of misinformation spreading, leading to superior predictive performance compared to purely graph-based or content-based approaches. Similarly, Qureshi et al. [25] propose integrating features derived from the interaction graph rather than using the graph directly within the neural network. These derived features reportedly exhibit strong discriminative power and are highly informative, while also improving computational efficiency compared to models that models that incorporate the graph structure within the neural network (e.g., through graph convolution or message-passing layers) during training and inference. Finally, Tommasel et al. [33] propose a three-component model that captures a user's tendency to spread misinformation by combining information from the publication interaction graph, the user interaction graph, and a set of textual and behavioral features.

Despite their strong empirical performance, existing hybrid approaches also present notable limitations. Many rely on complex, multi-stage pipelines and heavyweight architectures, which increase computational cost and hinder scalability and reproducibility. Moreover, feature fusion strategies are often tightly coupled to specific datasets or domains, limiting their generalization to other settings. In addition, most hybrid models do not explicitly incorporate temporal information, even though the dynamics of user behavior and information diffusion are central to misinformation spreading. These limitations motivate our hybrid approach, which follows a more lightweight and modular design to balance representational power and efficiency while supporting broader applicability and temporal analysis.

2.4 Temporal Encoding

Incorporating temporal information into misinformation detection models enables a more dynamic understanding of how user behavior and information diffusion evolve over time. Several approaches leverage sequential models to capture such temporal dependencies. For instance, Plepi et al. [23] employ a Gated Recurrent Unit (GRU) architecture to model temporal patterns by organizing training data into monthly batches and processing content in chronological order. This structured setup allows the model to learn the temporal evolution of misinformation-related behavior. However, such approaches typically require large amounts of training data to learn meaningful temporal representations.

Alternative recurrent architectures, such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, have also been explored for temporal encoding. Ghanem et al. [7] propose an LSTM-based model that processes time chunks constructed in a similar manner and incorporates an attention mechanism to focus on the most relevant tweets, thereby enhancing the learned representations. The authors argue that temporal modeling is beneficial, as users' exposure to news and their propensity to spread misinformation may change over time. Liu and Wu [15] propose an early misinformation detection framework comprising four components: propagation path construction, RNN-based propagation path representation, CNN-based propagation path representation, and

final classification. In their approach, user interactions are organized as a multivariate time series, and a Gated Recurrent Unit (GRU) [4] is employed within the RNN module to encode temporal-linguistic features and model temporal dependencies along the propagation path, thereby incorporating temporal information into the prediction process.

Chapter 3

Data and Data Preprocessing

3.1 Data Collections

Several data sources are available for this task, each containing different types of information. These collections vary not only in the nature of the data they include but also in the process of annotating ground truth labels, which typically requires manual verification. In this work, we will only discuss the following data collections, which are presented in Table 3.1.

FibVid. The FibVid dataset [13], collected during the COVID-19 period, comprises news claims extracted from several fact-checking journalism platforms, including *PolitiFact*[1] and *Snopes*[2] . It covers both COVID-19–related and contemporaneous non-COVID topics from January 2020 to December 2020. For each news claim, relevant keywords were extracted and used to query *Twitter* in order to retrieve associated content. The dataset contains 772 news claims associated with a total of 161,838 related tweets. Among these claims, 26% are labeled as authentic and 74% as fake. Tweets are labeled individually. In addition to claim-level information, the dataset provides propagation-related metadata, including retweet counts, post text, hashtags, like counts, and creation dates, as well as user-level information, including account creation date, profile description, follower count, and following count. The final collection comprises 24,430 users and 296,804 tweets.

Pheme [38]. It consists of rumor and non-rumor conversation threads from *Twitter*, collected during major events such as the Ferguson unrest, the shooting at Charlie Hebdo, the shooting in Ottawa, the Sydney Siege, and the crash of a Germanwings plane. Each thread is initially annotated as either a rumor or a non-rumor based on human judgments; subsequently, rumor threads are further labeled as true, false, or unverified by professional journalists. Each conversation thread is structured as a tree, originating from a source tweet and followed by a series of responses. The total number of tweets in this dataset is 105,354.

PAN 2020 [1]. It consists of 20,000 English and 20,000 Spanish anonymous tweets. Unlike the other collections, it includes only the textual content of users' tweets, without access to conversation threads, interactions, or any additional user metadata.

In this work, we primarily rely on the *FibVid* dataset due to its large volume of verified data and the flexibility of the available information, which enables the construction of additional features. The dataset provides rich user-level attributes, supporting a detailed analysis of user behavior and model performance. Although the data are centered on the COVID-19 period, the dataset covers a variety of sub-topics within this broader event, allowing for analyses across different narratives and claims. Although the data are centered

[1] https://www.politifact.com/
[2] https://www.snopes.com/

Dataset	#Tweets	Language	Tree of Responses	Additional User Info	Labels
FibVid	161,838	English	Yes	Yes	per tweet
Pheme	105,354	English	Yes	Yes	per tweet
PAN 2020	20,000 + 20,000	English/Spanish	No	No	per user

Table 3.1: Comparison of the analyzed data collections and their features.

on the COVID-19 period, the dataset covers a variety of sub-topics within this broader event, enabling analyses across different narratives and claims. This provides a rich and diverse setting within a single real-world context, allowing for a focused and controlled evaluation of model behavior. However, because all data stem from the same overarching event, the dataset might not support cross-event generalization, which constitutes an inherent limitation.

3.2 Linguistic Features

To account for the heterogeneous nature of user representations on social media, our approach adopts feature extraction strategies inspired by prior work on misinformation, bias, and rumor detection. Prior work has shown that these features can effectively distinguish misinformation spreaders from regular users even without relying on graph-based models such as Graph Convolutional Networks (GCNs) [14]. Nevertheless, combining these features with graph-based information can yield more robust and accurate predictions, as further investigated in our ablation study. The following features are extracted to construct the user feature vector:

1. *User metadata.* This category includes statistics describing the user profile, such as the follower-to-friend ratio, description length, whether the user uses a default profile picture, the number of days since account creation, name length and digit count, as well as the numbers of statuses, favorites, followers, friends, and list memberships.

2. *User writing behavior.* This category captures user-specific writing patterns in user-generated content, including the ratio of words to characters, the relative frequency of uppercase and lowercase words, and the average sentence length. Additionally, given the widespread use of emojis in social media communication, emoji usage is included and quantified as the number of emojis normalized by text length.

3. *Tweet statistics.* This category comprises general statistics derived from tweet activity, including the proportion of tweets containing mentions, URLs, or hashtags; the proportion of replies; an insomnia index defined as the percentage of tweets posted at night; the proportion of tweets posted on weekends; tweet and reply counts; favorite counts and favorite ratios; retweet counts and retweet ratios; and the average tweet length.

4. *Punctuation.* Following prior work (e.g., [8, 22]), punctuation-based features are included, as they have been shown to differentiate between deceptive and truthful texts. Although these findings stem from text-level analyses, they may also be indicative of user-level patterns when aggregated across publications. Specifically, we consider the frequency of question marks, exclamation marks, periods, commas, dashes, colons, double question marks, double exclamation marks, semicolons, and ellipses.

5. *LIWC categories.* We extract a subset of linguistic features from the LIWC lexicon

[19], focusing on categories related to personal pronouns, personal concerns, cognitive processes, informal language, and perceptual processes.

6. *Tree metrics.* This category includes features derived from the interaction graph, a network of users connected through their interactions, such as tree height and size, the average similarity between nodes in a tree based on BERT representations of the corresponding publications, the reciprocity of interactions within the tree, and the average reply time in a conversation thread, defined as the time elapsed between the original post and the reply by the target user. Similar metrics have been shown to outperform alternative baselines in prior work [32].

7. *Content evolution and user publication behavior* (hereafter *Content evolution*). This category captures how a user's content and interaction patterns change over time. It includes the average semantic similarity between the user's first and last publications, the reciprocity of user interactions, measuring the degree of mutual exchange in conversations, the h-index of the user's conversation thread reflecting participation in sustained discussion threads, and the semantic similarity between the user's first and last publications and the average semantic representation of their overall content. These features are inspired by prior work [32].

8. *Readability scores.* This category comprises metrics that estimate the structural complexity of a text and the level of literacy required to understand it. Prior work by Pérez-Rosas et al. [22] showed that readability features can outperform LIWC and other feature types, such as n-grams, punctuation, psycholinguistic features, and syntactic features, in misinformation detection within their experimental setting. Readability scores are computed following the methodology of Shrestha and Spezzano [29]. We use the following metrics: Flesch Reading Ease, Flesch–Kincaid Grade Level, Coleman–Liau Index, Gunning Fog Index, SMOG Index, Automated Readability Index, and Dale–Chall Readability Score.

9. *Personality traits.* We model user personality using the five-factor (OCEAN) model, which comprises Extroversion, Neuroticism, Agreeableness, Conscientiousness, and Openness [17]. Personality traits are inferred from users' textual content using a publicly available BERT-based model[3] fine-tuned for personality prediction, where BERT provides contextualized text representations [5]. Such features have been widely used in recommendation systems [28, 36] and have also been applied in fake news and misinformation detection [2, 8, 29].

10. *Emotion and sentiment.* Misinformation, particularly when spread intentionally, often aims to elicit strong emotional reactions such as outrage[4]. Consequently, emotional features are widely used in prior work (e.g., [7, 8, 29]). Following the Plutchik model [24], we consider the core emotions fear, anger, trust, surprise, sadness, disgust, and joy. In addition, sentiment is computed using the NRC Emotion Lexicon [18], motivated by the hypothesis that misinformation spreaders and regular users differ in their use of sentiment-laden language.

11. *Morality scores.* This category captures moral framing using features derived from Moral Foundations Theory (MFT) [9], which describes moral reasoning along five foundations: harm, fairness, ingroup (loyalty), authority, and purity (sanctity). Each foundation is further divided into virtue (positive) and vice (negative) dimensions, along with an overall morality score. Moral values have been discussed in prior studies

[3] https://huggingface.co/Minej/bert-base-personality
[4] https://hsph.harvard.edu/health-communication/news/lessons-learned-to-fight-misinformation-focus-on-emotions/

as influencing misinformation dissemination[5]. The scores are computed following the approach described in [7].

3.3 Exploratory Feature Analysis

We examine the discriminative power of newly proposed features, in particular, morality, emotions, and sentiment, in terms of how well they characterize the spreaders and non-spreaders groups. Differences between spreaders and non-spreaders were evaluated for each feature using Welch's t-test [35] to determine if two populations have different means. Significance is indicated on figures with asterisks ($p < 0.05$, $*p < 0.01$, $**p < 0.001$).

Figure 3.1 shows the distributions of selected lexical features and activity-related attributes. The average proportion of uppercase words per sentence exhibits very similar distributions for both classes, with regular users showing a slightly higher mean value than spreaders (0.026 vs. 0.025, respectively). A similar pattern can be observed for the average word length. The box plot of emoji usage reveals only small visible differences; therefore, we additionally report the mean values, which indicate that regular users use emojis somewhat more frequently (mean = 2.26 in the unprocessed data) than misinformation spreaders (mean = 2.10 in the unprocessed data). In contrast, the features describing user activity show more noticeable differences between the two groups. The nocturnal activity feature displays an approximately 10% difference in the median, while the mean values remain close (0.41 vs. 0.42 for non-spreaders and spreaders, respectively). Finally, the feature capturing the proportion of publications posted on weekends exhibits the clearest difference in the observed distributions.

Figure 3.2 shows the distributions of the Big Five personality traits for the two user groups. Several traits display visible differences in their distributions, while others appear more similar across classes. In particular, Neuroticism shows relatively similar distributions for regular users and misinformation spreaders compared to the other traits.

Figure 3.3 presents the distributions of emotion-related features. For some emotions, such as Sadness, Surprise, and Anger, the distributions across the two groups appear largely similar. In contrast, features such as Anticipation and Disgust exhibit more noticeable differences in their observed distributions.

[5] https://www.msn.com/en-in/lifestyle/smart-living/study-highlights-connection-between-your-personal-values-and-fake-news/ar-AA1GNvbJ

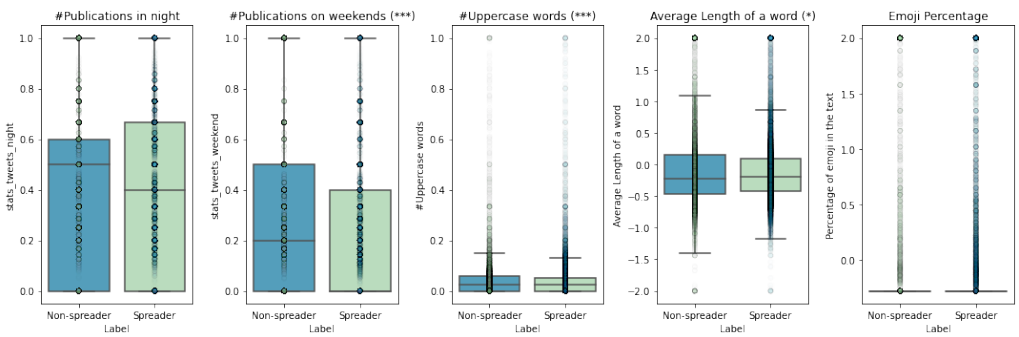

Figure 3.1: Distribution of selected lexical features and activity behavior for spreaders and non-spreaders. Significance is indicated with asterisks ($p < 0.05$, $*p < 0.01$, $**p < 0.001$).

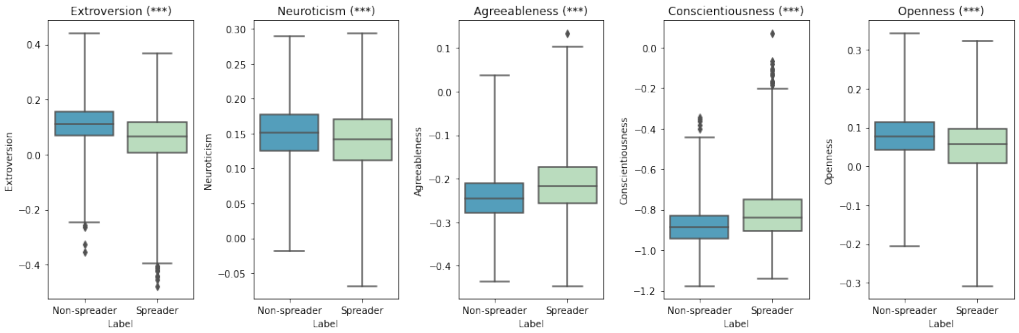

Figure 3.2: Distribution of OCEAN personality traits [17] for spreaders and non-spreaders. Significance is indicated with asterisks ($p < 0.05$, $*p < 0.01$, $**p < 0.001$).

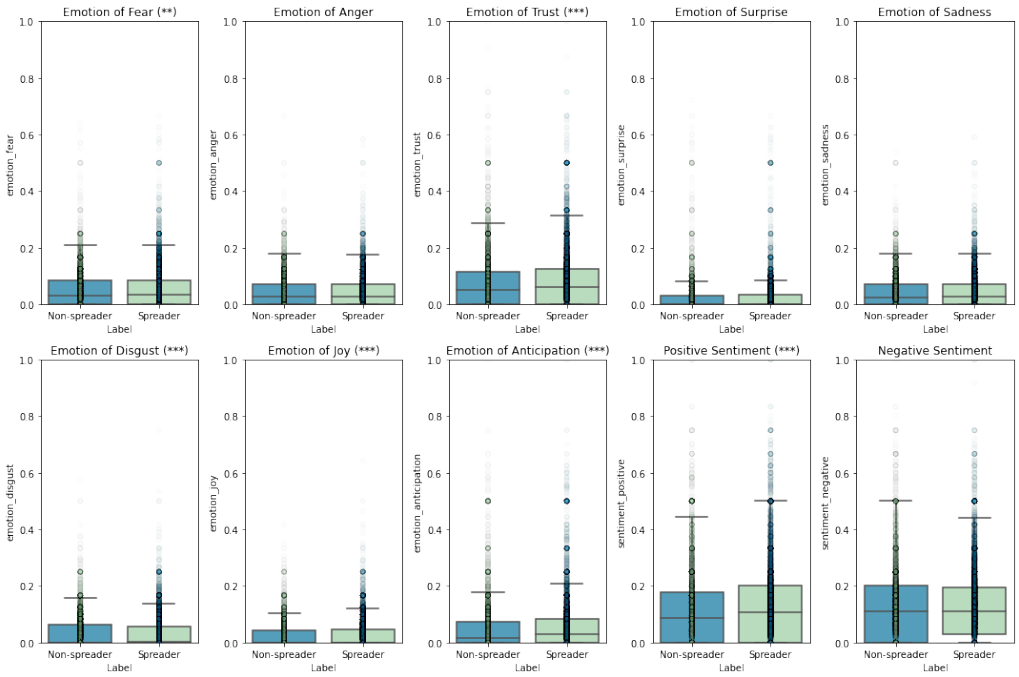

Figure 3.3: Distribution of emotions from the NRC Emotion Lexicon [18] for spreaders and non-spreaders. Significance is indicated with asterisks ($p < 0.05$, $*p < 0.01$, $**p < 0.001$).

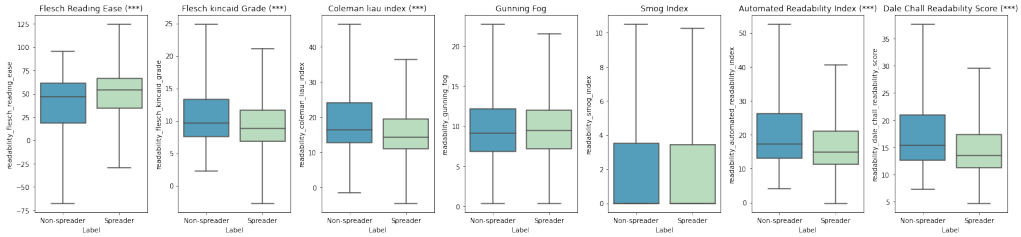

Figure 3.4: Distribution of various readability scores for spreaders and non-spreaders. Significance is indicated with asterisks ($p < 0.05$, $*p < 0.01$, $**p < 0.001$).

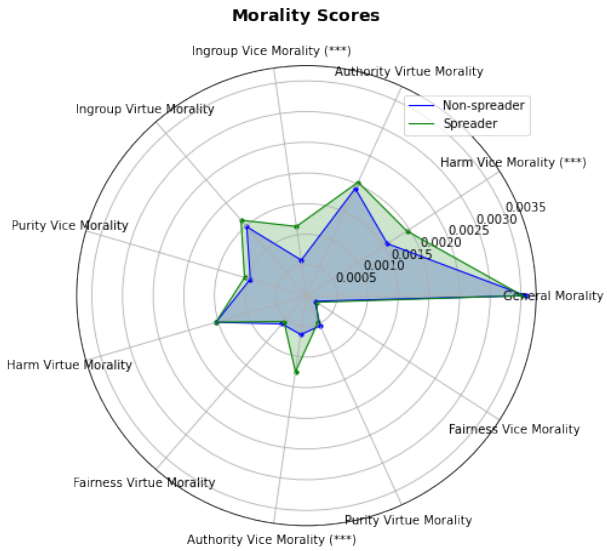

Figure 3.5: Distribution of morality scores for spreaders and non-spreaders. Significance is indicated with asterisks ($p < 0.05$, $*p < 0.01$, $**p < 0.001$).

With respect to linguistic complexity, the readability metrics shown in Figure 3.4 reveal modest differences between the two groups. Among these, the Dale–Chall Readability Score and the Coleman–Liau Index show more pronounced distributional differences compared to the other readability measures.

To analyze moral foundations, we examine the morality scores using radar plots due to the fact that many features have median values of zero. Figure 3.5 indicates that the distributions of Ingroup Vice, Authority Vice, and Harm Vice differ more visibly between the two groups than the remaining moral dimensions.

Figure 3.6 summarizes user-level statistics. Some features show noticeable differences in their distributions, in particular the number of days since account creation. The feature describing the length of the user's screen name shows almost identical distributions across the two classes. In addition, non-spreaders are more often verified than spreaders (0.10 vs. 0.05, respectively) and are less likely to use a default profile picture (0.02 vs. 0.07, respectively).

Tree-based and content evolution metrics are shown in Figure 3.7. These features display clear distributional differences between the two groups. In particular, the propagation trees associated with misinformation spreaders tend to be larger and deeper, while the opposite trend is observed for non-spreaders. Moreover, the semantic similarity between the first and last publications shows different distributional patterns when computed

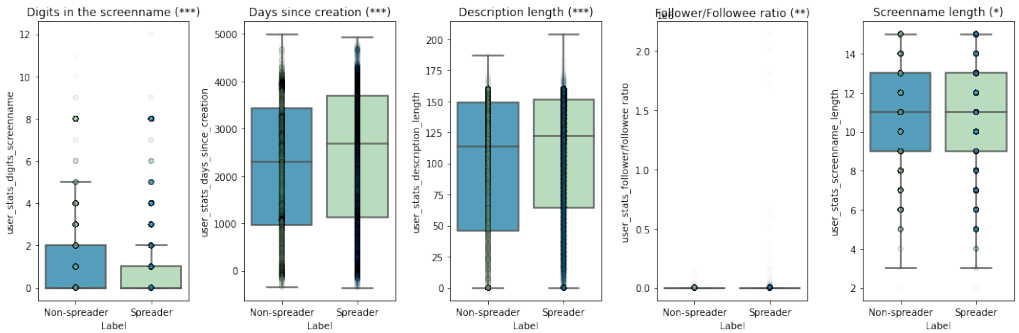

Figure 3.6: Distribution of user general statistics for spreaders and non-spreaders. Significance is indicated with asterisks ($p < 0.05$, $*p < 0.01$, $**p < 0.001$).

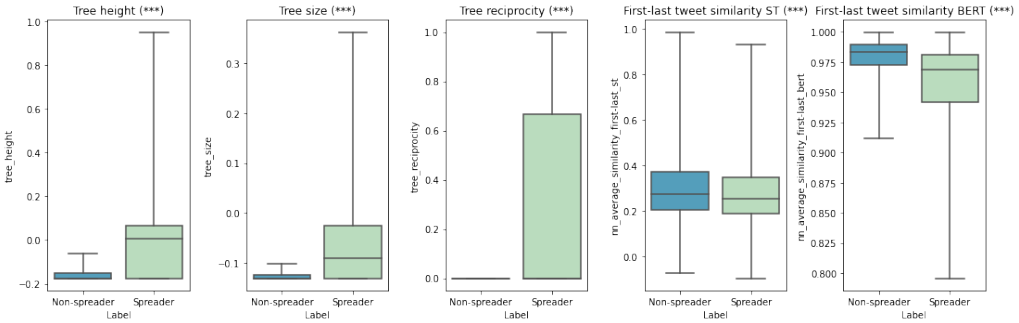

Figure 3.7: Distribution of tree metrics and content evolution metrics for spreaders and non-spreaders. Significance is indicated with asterisks ($p < 0.05$, $*p < 0.01$, $**p < 0.001$).

using BERT-based representations, whereas SentenceTransformer-based representations[6] yield more similar distributions across groups.

Figure 3.8 shows the class distributions of the tweet statistics features. Several attributes exhibit noticeable differences between the two groups, in particular the percentage of mentions in a user's tweets, the percentage of tweets containing URLs, and the favorites count.

In preliminary experiments, a baseline model trained using only these tweet statistics features achieved higher performance than the other tested models. An analysis of feature importance based on a random forest model indicated that the presence of a URL received the highest importance score (0.10), which was markedly higher than that of the remaining features.

However, the presence of a URL does not, by itself, indicate whether a post contains misinformation. The observed pattern is likely influenced by characteristics of the dataset, such as the inclusion of a large number of posts from official news sources, which the model may learn to associate with the target labels.

Due to the risk of bias introduced by the data collection process and the lack of independently collected data to validate these features, the tweet statistics feature group is excluded from the final evaluation in order to avoid potentially misleading or non-generalizable results.

[6] https://huggingface.co/sentence-transformers/all-mpnet-base-v2

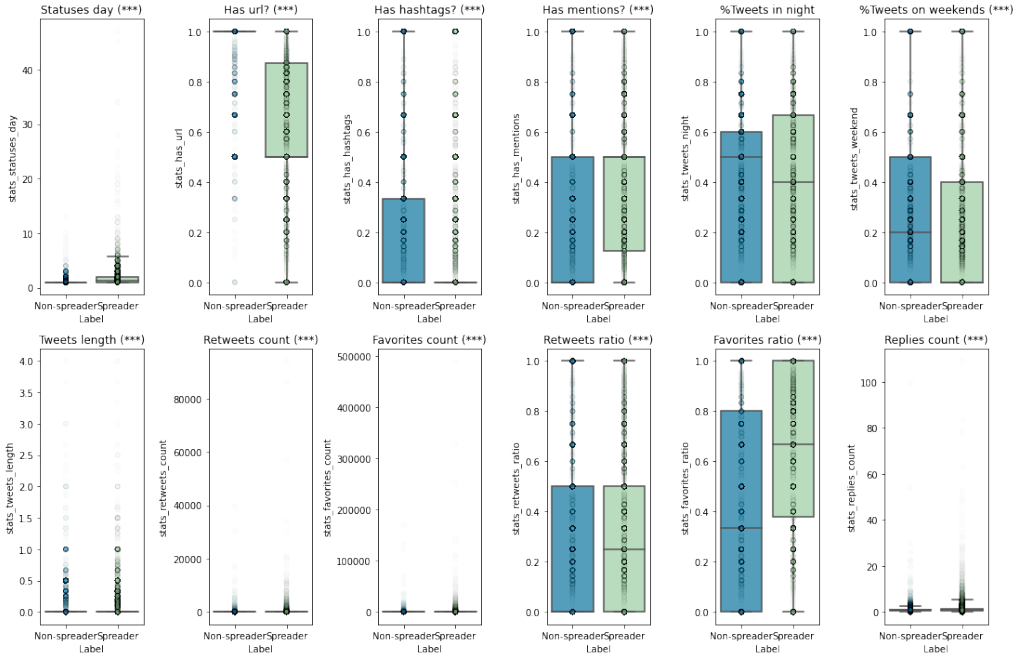

Figure 3.8: Distribution of tweet statistics for spreaders and non-spreaders. Those metrics were excluded during the training. Significance is indicated with asterisks ($p < 0.05$, $*p < 0.01$, $**p < 0.001$).

Overall, the demographic and user-content-derived features analyzed in this section exhibit observable differences between regular users and misinformation spreaders, motivating their inclusion in subsequent modeling experiments.

Chapter 4

Model

The proposed model follows a hybrid structure, integrating content-based features with graph-based representations to enhance the detection of misinformation spreaders. An overview of the model architecture is presented in Figure 4.1. The model consists of four parts, the results of which are concatenated, fed through the Feed-Forward Neural Network, and used for the classification. The model incorporates three graph-based parts: *User-with-User Similarity Graph*, *Publication-with-Publication Interaction Graph*, and *User-with-User Interaction Graph*, as well as a user feature set described in Section 3.2. The selected subset of user features is used as input to both the *User–User Similarity Graph* and the *User–User Interaction Graph*.

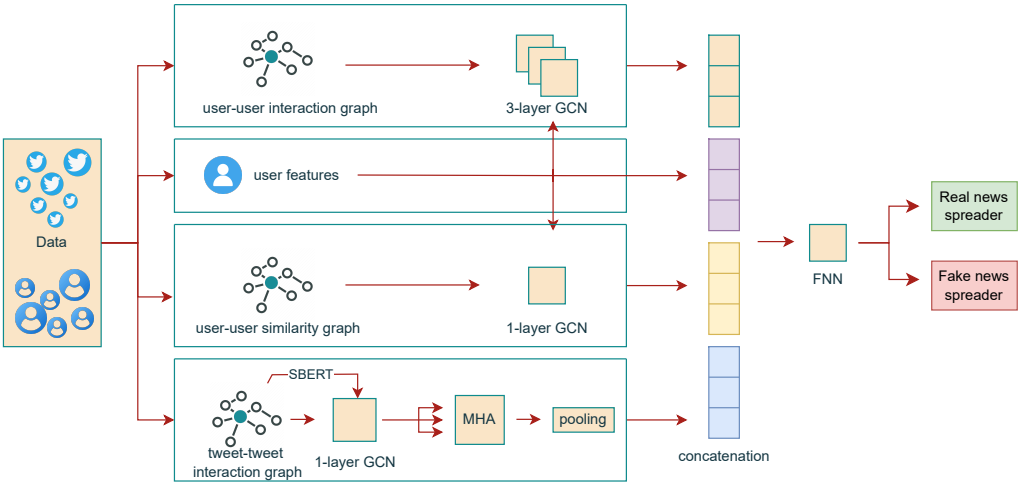

Figure 4.1: Pipeline overview of a proposed model

4.1 User-with-User Similarity Graph (UUSG)

Prior work by Plepi et al. [23] suggests that similarity graphs can outperform simple interaction graphs for misinformation spreader detection. Building on this observation, we integrate a similarity graph as a parallel component to the interaction graph in order to better capture relationships between users with similar posting behavior. However, since our model already incorporates semantic content information through the user-content-derived attributes described in section 3.2 and combines these with other representations for final classification, we construct the similarity graph using S-BERT embeddings to reduce redundancy in the captured relationships.

Semantic similarity between users is measured using cosine similarity, and an edge is created between two users if their similarity exceeds a predefined threshold Φ. Given the high degree of clustering in the data, some users may otherwise be connected to a very large number of similar users, which would increase computational cost and introduce redundant information. To address this issue, we limit each user's connections to the top k most similar nodes.

4.2 Temporal Encoding

Publications shared within a short time period are assumed to be semantically related, as they are likely to convey similar meanings and refer to the same events/topic. To achieve this, a one-day sliding window is applied, as this duration is expected to capture the emergence and discussion cycles of a news claim. Each aggregated publication group containing n publications is weighted by $\log(1 + n)$, assigning greater importance to larger groups while slightly discounting tweets posted within a short time span, under the assumption that they might convey redundant information. To prevent excessive aggregation, the maximum length is set to seven days, ensuring that publications are not indefinitely stacked into a single unit.

Additionally, we introduce a time-based weighting mechanism under the assumption that recent tweets are more indicative of user behavior. Each tweet cluster is weighted by a scaling factor a, which is computed using a scaled and shifted parabola applied to the tweet's timestamp. The value a is constrained to the interval $[0.5, 1]$, ensuring that older publications are not completely disregarded but are given lower significance. This time-aware weighting mechanism enables a more nuanced representation of evolving user behavior. The approach is conceptually aligned with the Normalized Discounted Cumulative Gain (NDCG) definition and captures finer-grained temporal dynamics compared to a simple mean aggregation of all features. This weighting is applied during user embedding construction, so that newer tweets contribute more strongly to the user representation, while older tweets are down-weighted but still included.

4.3 Publication-with-Publication Interaction Graph (PPIG)

To enrich the model's contextual information, we incorporate publication-conversation threads to extract contextual relationships. To achieve this, we first tokenize the conversation threads using BERT [6]. The tokenized representations are then processed through a single-layer GCN to capture structural dependencies within the conversation network. Then, the GCN output is used to generate Query, Key, and Value matrices that serve as input to a multi-head self-attention mechanism. This architectural design enables the model to uncover deeper, indirect interaction patterns among users, beyond their immediate, direct connections. Following the attention mechanism, we apply a pooling layer to consolidate the extracted information. The resulting vector, which represents the user's publication interactions, is then concatenated with the user feature vector, ensuring that both content-based and interaction-based information are integrated into the final user representation.

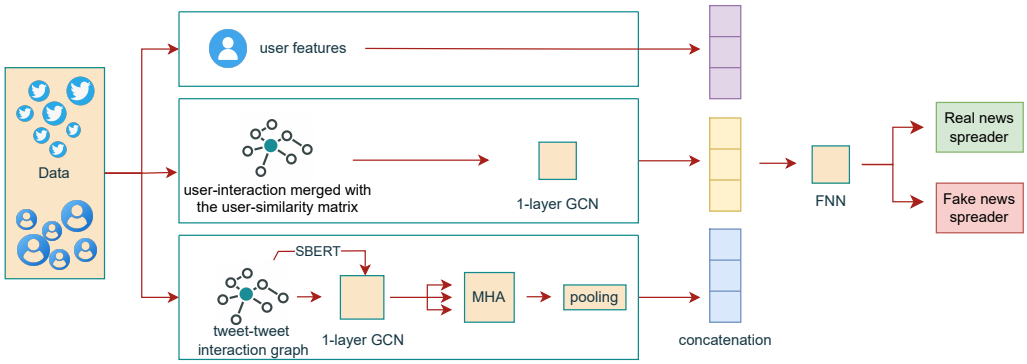

Figure 4.2: Pipeline overview of a User-with-User Similarity-Interaction-United Mode (SIUM).

4.4 User-with-User Interaction Graph (UUIG)

An interaction is defined as any engagement between users through replies, mentions, or responses to published content. The model is agnostic to the specific type of interaction: any such engagement between two users is treated as an interaction. Based on this definition, we construct an interaction graph that captures both direct and indirect user interactions, where indirect interactions are defined as implicit interactions occurring at a hop distance greater than two. Whenever a user's publication involves an interaction with another user, an edge is created between the corresponding users. This graph representation enables the model to capture patterns of user interactions, which have been shown to be effective in prior work [12, 26, 32].

4.5 User-with-User Similarity-Interaction-United Model (SIUM)

In addition to the primary model, a second model is proposed that integrates the user interaction matrix and the user similarity matrix into a single representation. The processing pipeline for this model is illustrated in Figure 4.2.

SIUM combines the user interaction and user similarity matrices using a logical OR operation and applies a single GCN to the resulting matrix, following the same pipeline as the primary model. By operating on a unified User-with-User Similarity Graph and avoiding the use of parallel GCNs, SIUM reduces computational complexity, achieving an approximate 40% improvement in efficiency in the experimental results.

This model is included in the comparative evaluation and the ablation study to assess the trade-offs between representational expressiveness and computational efficiency when jointly modeling user interactions and user similarities.

4.6 Model Implementation Details

As a preprocessing step, extreme feature values are clipped to the range $[-2, 2]$ to reduce noise and improve training efficiency [37]. To model social interactions, we employ Graph Convolutional Networks (GCNs) [14], a commonly used and powerful graph embedding

technique that captures both local graph structures and node features. The model utilizes three concatenated GCN layers, enabling the incorporation of information related to up to 3-hop neighbors for a *User-with-User Interaction Graph*, two concatenated GCN layers for a *Publication-with-Publication Interaction Graph*, and one GCN layer for a *User-with-User Similarity Graph* due to its high density. This approach ensures that both users' direct and indirect interactions are accounted for, providing a richer representation of their network activity. GCNs were implemented as follows:

$$\mathrm{GCN}(X, \hat{D}, \hat{A}) = f\left(\hat{D}^{-\frac{1}{2}}\hat{A}\hat{D}^{-\frac{1}{2}}XW + b\right)$$

Where A represents the adjacency user matrix derived from the user interaction graph, and I represents the identity matrix. The normalized adjacency matrix is defined as $\hat{A} = A + I$, and $\hat{D}_{ii} = \sum_j \hat{A}_i j$, where f is the activation function (e.g., ReLU), W is the trainable weight matrix, b is the trainable bias vector, and X represents the user feature matrix, which incorporates both user-specific features and those of their neighbors, as defined by A. The features described in Section 3.2 are used to construct X. The list of features fed to the GCN is a hyperparameter. We pass the user interaction graph through the three layers of the GCN, producing an output matrix that encodes representations for all users in X. Finally, only the vector corresponding to the target user is retained for classification.

GCNs are applied in parallel to the *User-with-User Similarity Graph*, *User-with-User Interaction Graph*, and *Publication-with-Publication Interaction Graph*. The resulting vectors from each part of the model are concatenated with one another and with the full set of user-specific features. To retain user-specific signals not fully captured by graph structures, all user features are included in the final concatenation. A subset of these features is also used as the input feature matrix X for the GCNs. The resulting concatenated vector is passed through a Feedforward Neural Network (FNN) for final classification. The ReLU activation function is employed to introduce non-linearity and enhance model expressiveness. To ensure efficient training, data is processed in mini-batches of n users.

Chapter 5

Experimental evaluation

To evaluate the performance of the proposed model, we address the following research questions, which aim to provide a comprehensive assessment of the models effectiveness:

1. **RQ1.** How does the proposed model compare with state-of-the-art techniques in predicting the future behavior of unseen users?
2. **RQ2.** What is the contribution of each component of the proposed pipeline to the overall performance of the model?
3. **RQ3.** How does the proposed model perform in a multi-class prediction setting?

5.1 Experimental settings

Data partition. From the training set, a validation set is created by randomly selecting 10% of the users. Since the training and test sets do not share users, the task focuses on predicting whether previously unseen users (i.e., users unknown at training time) are misinformation spreaders. Unlike the traditional detection scenario, where features and structures are built based on complete propagation and interaction networks after publications have been spread (e.g., [7, 8, 12]), temporal partitions allow for modeling the sequence and timing of interactions over time. This approach not only captures interaction and propagation structures, but also enables the understanding of temporal dynamics. For consistency and comparability, all models are trained and evaluated on identical training, validation, and test splits.

Misinformation spreader definition. The definition of a misinformation spreader depends on the proportion of misinformation content that a user shares. A threshold of 0.5 is adopted, meaning that a user is defined as a spreader if more than half of their shared content is identified as misinformation. This choice represents a conservative and commonly used convention in prior research, striking a balance between including users who predominantly disseminate misinformation while avoiding overly inclusive or overly restrictive class definitions. Under this threshold, 42% of users are classified as non-spreaders and 57% as misinformation spreaders, yielding a reasonably balanced class distribution.

Multiclass definition. In addition to binary classification, a multiclass classification setting is considered to provide a more fine-grained characterization of user behavior. In this setting, multiple thresholds are used to partition the continuous misinformation ratio into discrete classes. Given m thresholds, this results in $m + 1$ classes defined as $[0, k_1), ..., (k_m, 1]$. This work focuses on a three-class configuration, which is sufficient

for common misinformation detection scenarios and introduces an intermediate category between non-spreaders and spreaders. Two thresholds, set at 0.35 and 0.65, are used to define these classes. Users with a misinformation ratio below 0.35 are classified as non-spreaders, users with a ratio above 0.65 are classified as misinformation spreaders, and users with ratios in between are categorized as potential misinformation spreaders. The intermediate class captures users whose behavior is less clearly defined and may benefit from further observation or additional contextual information, particularly in online or evolving settings. This three-class formulation therefore provides a more nuanced representation of user behavior while avoiding unnecessary complexity. The resulting class intervals are $[0, 0.35), [0.35, 0.65), (0.65, 1]$.

Evaluation metrics. For evaluation, the task is primarily treated as a binary classification problem. Accordingly, Precision, Recall, F1-score, and the Matthews Correlation Coefficient (MCC) [3] are used. MCC is particularly informative in the presence of class imbalance, which is the case in this setting. These metrics are commonly adopted and well suited to binary classification tasks. The model prioritizes recall over precision, as the primary objective is to identify as many misinformation spreaders as possible and to minimize the risk of failing to detect such cases. For the three-class classification setting, the same evaluation metrics are applied. Precision, Recall, and F1-score are computed using macro averaging to account for class imbalance and to ensure equal weighting of all classes.

5.2 Baselines

The performance of the proposed model is evaluated against several baseline methods. These comparison models are grouped into two categories: (1) simple baselines that incorporate only specific components of the proposed model, and (2) state-of-the-art models.

First, a set of feature-based baselines is evaluated, where each model considers a specific subset of features independently.

1. User stats (basic data regarding the user that was derived without considering tweet-content, e.g., follower/friend count, verification status, account age, description length, etc.)
2. LIWC categories (a subset of LIWC linguistic categories. LIWC relies on extensive lexicons of word categories that reflect underlying psycholinguistic processes.).
3. Readability (indices measuring the structural complexity of a text and the literacy level required to comprehend the information given).
4. Personality traits (big five personality traits, referred to as OCEAN [17], which are commonly used to understand and measure individual differences in personality).
5. Emotion and sentiment (the abstract attributes capturing emotionality derived from tweet content via the NRC Emotion Intensity Lexicon).
6. Tree metrics (features derived from the interaction graph, which provide an overview of tree size, height, and semantic similarity between publications in the tree).
7. Punctuation (The usage of punctuation by a user. This includes punctuation characters such as periods, commas, dashes, question marks, exclamation marks and others).

8. Tweet-content embedding (a 384-dimensional dense vector that were calculated using a SentenceTransformer architecture *sentence-transformers/all-MiniLM-L6-v2* [34]).
9. Content evolution (the average similarity between first and last publications, capturing the extent of how similar the opinion user is sharing over time).
10. Morality scores (features derived following Moral Foundations Theory (MFT), capturing an abstract description of the morality of users' publications).
11. All features (the combination of all features that will be used in the final model).

We evaluate several standard classifiers for the feature-based baselines, including Gradient Boosting, Logistic Regression, Naive Bayes, Decision Tree, Feedforward Neural Network (FNN), Support Vector Machine (SVM), k-Nearest Neighbors (KNN), and Random Forest. For each feature set, the best-performing classifier is selected based on the Matthews Correlation Coefficient (MCC) measured on the validation set. For every method, we conduct a simple grid search over hyperparameters using the validation set for optimization. When available, we use the original implementations of baselines from related work, and hyperparameter tuning follows the procedures described in the corresponding studies.

State-of-the-art baselines:

1. Hamdi et al. [11], based on hand-crafted user and node2vec features [10], that are processed through the pipeline and traditional classifiers.
2. *CheckerOrSpreader* [8], based on psycho-linguistic features and BERT embeddings [6], that are fed to the Convolutional Neural Network for the classification. The original model utilizes a "communication style" feature set to describe the intentions behind users' posts. These features could not be reproduced, as the feature extraction process is not publicly available.
3. Tommasel et al. [33], based on uniting user features with content, incorporating GNN for both.
4. *FacTweet* [7], based on psycho-linguistic features and GloVe embeddings [20] processed through a recurrent neural network.

5.3 Implementation Details

Our proposed model, related work baselines, and simple baselines that consider a specific feature set were implemented in Python using *sklearn* and *PyTorch*. For the related work models, *TensorFlow* was additionally used. All visualizations were implemented with the support of *matplotlib.pyplot*. For the Sentence-Transformers and S-BERT, we used Hugging Face public models[1].

The size of the *User-with-User Interaction Graph* GCN was set to 32 for all three layers, i.e., $[32, 32, 32]$. The size of the *User-with-User Similarity Graph* GCN was set to 32 with 1 GCN layer, i.e., $[32]$. The size of the *Publication-with-Publication Interaction Graph* GCN was set to 64 and 16, with 2 layers, i.e., $[64, 16]$, using Multi-Head Attention (MHA), where the size of each head is set to 32 with 2 heads. For the final user features set, which is concatenated with the results of all GCNs, we include all features. Tree metric features are used as the input X for the GCNs operating on the *User-with-User Interaction Graph* and the *User-with-User Similarity Graph*. The architecture of the

[1] https://huggingface.co/models

final Feed Forward Neural Network we used was $[32, 64, 32]$ with other values being explored, but performed worse, via grid search in hyperparameter optimization (specifically $[32, 64, 32]$, $[64, 32]$, $[32, 32]$, $[128, 64]$).

The optimization for all reported results was performed using AdamW with a learning rate of (5×10^{-4}), and parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$, with other learning rates considered, such as $[(1 \times 10^{-3}), (1 \times 10^{-4})]$. We used HingeLoss as it is well-suited for margin-based binary classification, encouraging a clear separation between classes. We also explored BCELoss. The maximum number of epochs was set to 20, but all models converged in approximately 13 epochs; therefore, all results are based on models trained for 13 epochs. For the SIUM variant, training was set to 10 epochs due to its simplified architecture, which also enables faster training. All of the training was done in batches of 12, with other batch sizes explored in hyperparameter optimization, which performed worse, such as $[16, 32, 64]$. All hyperparameters were optimized through a simple grid hyperparameter search.

As noted earlier, semantic similarity is measured using cosine similarity, and an edge is created only if the similarity exceeds a predefined threshold Φ. Unless stated otherwise, we set $\Phi = 0.67$ in our experiments. To avoid overly dense neighborhoods for highly connected nodes, we retain only the top k most similar users per node, with k set to 150 unless otherwise specified. All hyperparameters are selected based on performance on the validation set.

To contextualize the reported training time, experiments were conducted on an Acer Predator PH-315-52 equipped with an Intel i7-9750H CPU and an NVIDIA GeForce GTX 1660 Ti GPU. Each training epoch takes ~ 79 seconds, and user classification takes ~ 0.06 seconds.

5.4 Evaluation

RQ1. *How does the proposed model compare with state-of-the-art techniques in predicting the future behavior of unseen users?*

Table 5.1 reports the performance of all evaluated methods. To ensure the accuracy and estimate the distribution of an estimator, we applied a bootstrapping procedure to the test set to estimate 95% confidence intervals for each evaluation metric and investigate any overlaps. The proposed model achieves the highest F1-score (0.823 ± 0.012) and MCC (0.613 ± 0.023) among all approaches, while ranking second in recall (0.855 ± 0.027) and third in precision (0.794 ± 0.015). Overall, this indicates a well-balanced trade-off between precision and recall compared to the other models.

The bootstrapped comparisons indicate that the primary model outperforms all traditional attribute-based baselines as well as the considered state-of-the-art models. The mean differences in MCC between our model and these baselines range from 0.08 to 0.46 (excluding the SIUM model), with 95% confidence intervals that do not include zero.

The average relative improvements over the baselines were 323% (±531%) for Matthews Correlation Coefficient, 66% (±77%) for F1-score, 16% (±13%) for precision, and 121% (±150%) for recall. The confidence intervals obtained via bootstrap analysis indicate that these improvements are robust, highlighting the stability and reliability of the proposed approach.

Considering the simple feature-based baselines, combining all features via concatenation (without the proposed graph-based architecture) already yields competitive results, outperforming most related methods and all other simple baselines. Among the individual

Classifier	F1-score	Precision	Recall	Matthews
Our proposal	**0.823**	0.794	0.855	**0.613**
SIUM	0.807	0.782	0.83	0.58
all features	0.767	0.801	0.736	0.532
content evolution	0.793	0.687	**0.938**	0.515
LIWC categories	0.496	0.593	0.427	0.104
punctuation	0.535	0.721	0.425	0.262
readability	0.453	0.669	0.342	0.173
tree metrics	0.741	0.67	0.828	0.392
user-stats	0.412	0.693	0.293	0.178
emotion	0.251	0.559	0.162	0.028
personality	0.601	0.759	0.499	0.338
morality	0.229	0.596	0.141	0.052
tweet-content embedding	0.767	0.767	0.736	0.538
Tommasel et al. [33]	0.76	**0.806**	0.72	0.527
CheckerOrSpreader [8]	0.655	0.79	0.576	0.381
Hamdi et al. [11]	0.725	0.674	0.784	0.371
FacTweet [7]	0.416	0.656	0.304	0.148

Table 5.1: Performance comparison across classifiers based on four evaluation metrics. The highest performance is written in bold, and the second-best is underlined.

feature sets, content evolution achieves the highest F1-score, even surpassing the full feature combination. It also attains the highest recall while maintaining relatively high precision, highlighting the usefulness of content-based embeddings in this dataset. This is consistent with the strong standalone performance of the SentenceTransformer-based representation, which achieves the third-highest MCC among all evaluated models.

The personality traits feature set performs better than the emotion and morality features in this setting, suggesting that personality-related signals capture different aspects of user behavior than emotional or moral cues. Tree-based metrics also achieve comparatively strong results, further supporting the relevance of features derived from content representations and interaction structure. In contrast, the user statistics feature set attains high precision but comparatively low recall, indicating that while profile-level information can be informative, it is insufficient on its own to capture the full range of misinformation spreading behavior.

State-of-the-art models generally achieve strong performance compared to the simple baselines. In particular, the model by Hamdi et al. [11] achieves the third-highest recall, albeit with lower precision. By contrast, the approach proposed by Tommasel et al. [33] attains the highest precision, but exhibits relatively low recall, reflecting a different trade-off between these metrics.

The lightweight model, SIUM, requires an average of 56 seconds per training epoch, compared to 79 seconds for the primary model, corresponding to a 41% reduction in training time. In terms of inference, the primary model takes on average 0.06 seconds to classify a user, whereas SIUM reduces this to 0.042 seconds, representing a 43% decrease in inference time.

RQ2. *What is the contribution of each component of the proposed pipeline to the overall performance of the model?*

To further investigate and determine the contribution of each component in the proposed pipeline, we conducted an ablation study. We will also discuss how some hyperparameters affect the model's performance. Our full proposal corresponds to the configuration described in section 5.3, using tree metrics as GCN input, all features concatenated at the user level, PPIG with two layers {64, 16}, UUIG with one layer {32}, AdamW optimizer, and Hinge loss. We considered the following model variants:

1. *Feature input* X *for a GCN (for User-with-User Interaction Graph and User-with-User Similarity Graph) - same vector input is given to both GCNs.* We evaluated the performance of our model with: (1) all features described in section 3.2, (2) content evolution and tree metrics features concatenated, (3) content evolution, tree metrics, and user statistics concatenated, (4) user statistics and tree metrics concatenated.

2. *User features concatenated to the output of the GCN that the FNN later treats.* We evaluated the performance of our model with: (1) no user features concatenated, (2) all features described in section 3.2, (3) content evolution and tree metrics features concatenated, (4) content evolution, tree metrics, and user statistics concatenated, (5) user statistics and tree metrics concatenated.

3. *Publication-with-Publication Interaction Graph GCN architecture.* We evaluated the performance of our model with: (1) conversation thread not being used directly, (2) PPIG GCN consists of three layers {16, 32, 16}, (3) PPIG GCN consists of two layers {32, 32}, (4) PPIG GCN consists of two layers {64, 64}, (5) PPIG GCN consists of one layer {100}.

4. *User-with-User Interaction Graph GCN architecture.* We evaluated the performance of our model with: (1) UUIG is not included in the model, (2) UUIG GCN consists of two layers {64, 32}, (3) UUIG GCN consists of three layers {64, 32, 16}.

5. *User-with-User Similarity Graph GCN architecture.* We evaluated the performance of our model with: (1) UUSG is not being used directly in the model, (2) UUSG GCN consists of one layer {64}, (3) UUSG GCN consists of one layer {100}.

6. *Loss function variant.* We evaluate a model variant using the Binary Cross-Entropy loss, while the full model employs the Hinge loss.

7. *Optimizer variant.* We also consider a model variant that uses the Adam optimizer instead of the default optimizer.

Table 5.2 summarizes the results of the ablation study. The full model achieves the best performance across all metrics except precision, where certain ablated variants obtain higher values. Overall, the ablated models tend to exhibit higher precision at the expense of lower recall. This trade-off is particularly visible when removing the Publication-with-Publication Interaction Graph GCN, which increases precision from 0.794 to 0.822. Similarly, removing the User-with-User Similarity Graph increases precision from 0.794 to 0.825, while removing the User-with-User Interaction Graph has only a marginal effect. In contrast, removing the user feature component leads to a consistent degradation across

	F1-score	Precision	Recall	Matthews
Our full proposal	**0.823**	0.794	**0.855**	**0.613**
feature input X for a GCN (for UUIG and UUSG) all features	0.725	0.812	0.655	0.492
feature input X for a GCN (for UUIG and UUSG) content evolution and tree metrics	0.799	0.808	0.791	0.581
feature input X for a GCN (for UUIG and UUSG) content evolution, tree metrics, and user statistics	0.79	0.811	0.769	0.57
feature input X for a GCN (for UUIG and UUSG) user statistics and tree metrics	0.772	0.815	0.734	0.548
UserFeatures no user features	0.787	0.758	0.818	0.533
UserFeatures content evolution and tree metrics	0.774	0.776	0.772	0.524
UserFeatures tree metrics	0.739	0.805	0.683	0.501
UserFeatures content evolution, tree metrics, and user statistics	0.802	0.787	0.817	0.573
UserFeatures user statistics and tree metrics	0.768	0.801	0.737	0.533
Publication-with-Publication Interaction Graph GCN absent	0.768	<u>0.822</u>	0.72	0.547
Publication-with-Publication Interaction Graph GCN three GCN layers {16, 32, 16}	0.757	0.819	0.703	0.533
Publication-with-Publication Interaction Graph GCN two GCN layers {32, 32}	0.816	0.794	<u>0.84</u>	0.601
Publication-with-Publication Interaction Graph GCN two GCN layers {64, 64}	0.81	0.792	0.828	0.589
Publication-with-Publication Interaction Graph GCN one GCN layer {100}	0.816	0.803	0.828	<u>0.604</u>
User-with-User Interaction Graph GCN absent	0.786	0.793	0.78	0.553
User-with-User Interaction Graph GCN two GCN layers {64, 32}	0.8	0.815	0.786	0.587
User-with-User Interaction Graph GCN three GCN layers {64, 32, 16}	0.782	0.805	0.76	0.554
User-with-User Similarity Graph GCN absent	0.744	**0.825**	0.678	0.522
User-with-User Similarity Graph GCN one GCN layer {64}	0.802	0.802	0.801	0.582
User-with-User Similarity Graph GCN one GCN layer {100}	0.816	0.797	0.835	0.601
TrainingLoss BCE	0.814	0.79	0.839	0.594
Optimizer Adam	0.808	0.796	0.821	0.589

Table 5.2: Ablation study results of a proposed model. The highest performance is written in bold, and the second-best is underlined.

threshold	k-top-sim	F1-score	Precision	Recall	Matthews
0.85	15	**0.807**	0.782	**0.834**	**0.58**
0.80	10	0.799	<u>0.805</u>	0.794	<u>0.579</u>
0.85	3	<u>0.801</u>	0.795	<u>0.807</u>	0.576
0.85	10	0.786	0.803	0.769	0.559
0.88	4	0.789	**0.817**	0.762	0.572

Table 5.3: Lighter model SIUM results (model described in section 4.5). The highest performance is written in bold, and the second-best is underlined.

all metrics. In these cases, the reductions in recall and Matthews Correlation Coefficient outweigh the gains in precision, resulting in lower F1-scores.

The full model benefits from the inclusion of all architectural components, as reflected in its higher recall and Matthews Correlation Coefficient. This indicates that integrating multiple information sources allows the model to capture a larger number of positive instances, albeit with a corresponding reduction in precision.

Replacing the Hinge loss with Binary Cross-Entropy (BCE) leads to a slight decrease across all metrics. A similar pattern is observed when using Adam instead of AdamW as the optimizer. Moreover, ablating the user features shows that reducing the number of input features negatively affects performance, suggesting that the full feature set contributes to the model's effectiveness. By contrast, increasing the number of attributes provided as input X to the GCNs (for both UUIG and UUSG) results in a performance decrease. Finally, variations in individual hyperparameters produce only modest changes in performance.

Table 5.3 reports the impact of the similarity threshold and the k-top-sim parameter on the SIUM model. For each configuration, only the GCN architecture is tuned based on validation performance, while the remaining hyperparameters are inherited from the full model. The configuration with a threshold of 0.85 and k-top-sim set to 15 yields the best results across all metrics except precision. The highest precision is achieved with a threshold of 0.88 and k-top-sim set to 4.

RQ3. *How does the proposed model perform in a multi-class prediction setting?*

The performance of the model in the multi-class prediction setting is reported in Table 5.4. In this evaluation, the multi-class setting is compared against simple baselines that incorporate specific components of the proposed approach, and against state-of-the-art models. The latter were, however, designed exclusively for binary classification tasks; essential editing was performed to adapt the models from related works to the multi-class setting. The choice of the loss function for the state-of-the-art models is Cross-Entropy loss.

The proposed models achieve the best scores in terms of F1-score, recall, and MCC. In terms of precision, our model ranks second, behind the classifier that uses all features combined, which places second across all metrics except precision. In the corresponding binary setting—where samples within the gray interval $[0.35, 0.65)$ are excluded—the final model attains precision and recall scores of 0.775 and 0.942, respectively.

Classifier	F1-score	Precision	Recall	Matthews
Our proposal	**0.659**	0.692	**0.645**	**0.542**
all features	0.622	**0.719**	0.602	0.525
content evolution	0.561	0.69	0.543	0.408
LIWC categories	0.295	0.352	0.365	0.105
punctuation	0.404	0.39	0.431	0.247
readability	0.373	0.377	0.409	0.198
tree metrics	0.463	0.557	0.463	0.301
user-stats	0.339	0.361	0.384	0.143
emotion	0.325	0.374	0.364	0.081
personality	0.437	0.418	0.465	0.329
morality	0.279	0.39	0.355	0.052
tweet-content embedding	0.492	0.467	0.52	0.469
Tommasel et al. [33]	0.588	0.616	0.584	0.41
CheckerOrSpreader [8]	0.521	0.526	0.52	0.404
Hamdi et al. [11]	0.403	0.391	0.428	0.265
FacTweet [7]	0.36	0.346	0.383	0.127

Table 5.4: Performance comparison across classifiers for 3-class classification. The highest performance is written in bold, and the second-best is underlined.

A closer inspection of the gray interval reveals that it contains 559 samples, of which 86% are non-spreaders and 14% are spreaders. This indicates that the model tends to assign borderline spreaders to the spreader class, exhibiting recall-oriented behavior, and leaves most intermediate samples in the gray interval. This way, the model maintains high recall for spreaders while preserving balanced performance. The binary setting, where we ignore the intermediate gray interval class and treat the initial class as non-spreaders and the last class as spreaders, yields slightly stronger overall results. The multi-class formulation provides a more fine-grained view of the model's predictions.

The modest performance improvement observed in the binary setting is expected, as excluding intermediate samples simplifies the classification task. The gray interval, however, captures cases associated with higher uncertainty and therefore remains an informative component of the framework, particularly in scenarios where it may be beneficial to delay final classification, for example, when a user's posting behavior is not yet conclusive. Users may benefit from continued observation as their class membership becomes clearer.

5.5 Summary of Results

1. **RQ1.** *How does the proposed model compare with state-of-the-art techniques in predicting the future behavior of unseen users?*

The proposed model consistently outperforms both state-of-the-art approaches and simple feature-based baselines across the main evaluation metrics. In particular, it achieves the highest F1-score and Matthews Correlation Coefficient, while also obtaining strong recall, indicating a well-balanced trade-off between precision and recall. These results suggest that the proposed architecture effectively integrates multiple sources of information to improve overall predictive performance.

2. **RQ2.** *What is the contribution of each component of the proposed pipeline to the overall performance of the model?*

   The ablation study shows that removing any major component of the model leads to a decrease in F1-score, recall, and MCC, highlighting the importance of each part of the pipeline. Among the components, the User-with-User Similarity Graph contributes most strongly to overall performance. While the removal of graph-based components leads to a slight increase in precision, it is accompanied by a notable decrease in recall and overall performance metrics, indicating that relational information plays a key role in capturing misinformation spreading behavior.

3. **RQ3.** *How does the proposed model perform in a multi-class prediction setting?*

   In the multi-class setting, the model achieves slightly higher overall performance compared to the binary classification scenario. This improvement is largely due to the exclusion of intermediate samples, which simplifies the classification task. An analysis of the class distributions shows that the intermediate (gray-interval) class is predominantly composed of non-spreaders, which helps explain the observed performance differences.

Chapter 6

Conclusion

The spread of misinformation has become a major societal challenge, contributing to increased anxiety and a decline in trust at individual, societal, and institutional levels. In this work, we focus on the task of detecting misinformation spreaders and systematically examine existing approaches by organizing prior research into three categories: content-based, graph-based, and hybrid models. Building on these lines of work, we propose a model that integrates multiple sources of information, including a user similarity graph, a user interaction graph, a tweet interaction graph, and user-derived features. The proposed approach leverages a graph neural network framework to learn joint representations from these heterogeneous sources. Experimental results indicate that the model effectively combines information from different modalities and outperforms the considered baselines. At the same time, due to the complexity of the full architecture, we introduce a simplified variant, SIUM, which achieves competitive performance and, in several metrics, outperforms the other evaluated models.

This study has several limitations that should be considered when interpreting the results. First, the proposed models are evaluated on a single dataset with a specific labeling procedure and predefined thresholds for identifying misinformation spreaders. Although this setup follows common practice in prior work, both the threshold selection and the labeling strategy may influence class distributions and model behavior. As a result, the generalizability of the findings to other datasets, platforms, or annotation schemes cannot be assumed. In addition, several feature groups—most notably the tweet statistics features—were excluded from the final evaluation due to biases introduced by the data collection process. While this decision improves the validity of the analysis, it also limits the exploration of potentially useful signals that might be applicable in settings with independently collected or more diverse data. Furthermore, the proposed hybrid model relies on multiple graph structures and extensive feature extraction, which increases computational complexity and data requirements. Although the SIUM variant partially mitigates this issue, the scalability of the full model to larger or more dynamic social networks remains an open question. Finally, the analysis focuses on observable user behavior and interaction patterns without explicitly modeling user intent or broader contextual factors, such as satire, irony, or coordinated campaigns. Consequently, the models are designed to identify patterns associated with misinformation spreading rather than to infer underlying motivations or causal mechanisms.

Several directions for future work emerge from this study. A primary avenue is the further simplification of the proposed architecture. While the hybrid model effectively integrates heterogeneous information sources, reducing the number of required graphs and feature sets (while preserving the most informative components) could improve scalability and facilitate deployment in real-world systems. The strong performance of the SIUM variant suggests that unified graph representations are a promising direction for achieving this

balance. Another important extension concerns the evaluation methodology. Future studies could employ multiple temporal splits, cross-platform datasets, or repeated training runs to enable statistical significance testing and to better assess the robustness of model performance across different settings. Additionally, evaluating the approach on datasets from domains beyond COVID-19–related misinformation would provide insights into its generalizability. The increasing availability of Large Language Models (LLMs) also opens new opportunities for misinformation spreader detection. LLMs could be leveraged to generate higher-level user representations, summarize behavioral patterns, or create auxiliary features that complement graph-based and linguistic signals. While such approaches may increase computational cost, they could enhance interpretability and support more transparent decision-making, particularly in high-stakes applications. Finally, future work could explore online and adaptive learning settings, where user behavior evolves over time and new data becomes continuously available. Incorporating mechanisms for incremental updates or concept drift detection may improve the model's ability to respond to changing misinformation strategies and emerging narratives.

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